300 measures of association, similarity, information and forecast evaluation for categorical data: the `catmetrics` command

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Abstract

This article introduces `catmetrics`, a new Stata command for computing measures of association, similarity, agreement, information, and deterministic and probabilistic forecast evaluation for categorical data. The command supports binary, nominal multicategory, and ordinal multicategory outcomes. It reports more than 300 measures, together with alternative names used across disciplines and information on their attainable ranges. In addition to overall measures, `catmetrics` computes class-specific quantities for each category and reports their macro and weighted averages.

Keywords: `catmetrics`, categorical variables, contingency table, confusion matrix, association, similarity, agreement, information, forecast evaluation, probabilistic forecast, categorical forecast.

1 Introduction

> ". . . there is no absolutely general measure of the degree of dependence. Every attempt to measure a conception like this by a single number must necessarily contain a certain amount of arbitrariness and suffer from certain inconveniences."
>
> — Cramér (1924)

Categorical data, including qualitative and discrete-valued numerical variables, are common across statistics and the behavioral, computer, decision, earth, engineering, information, life, medical, and social sciences. Over the past two centuries, researchers have proposed numerous measures to quantify association, similarity, agreement, dependence, information, and predictive performance for such data.

The resulting literature is rich but highly fragmented. Closely related quantities frequently appear under different names in different fields. Algebraically identical measures have often been independently rediscovered and renamed without recognition that they were already established elsewhere. As a result, the same measure may be used in meteorology, sociology, machine learning, epidemiology, and information retrieval under different labels and notations, with little or no cross-referencing.

Such reinvention is especially common when a measure addresses a universal methodological problem, such as the inflation of accuracy by unequal base rates. For instance, Cohen's kappa

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coefficient, introduced in 1960, is a widely used as a chance-corrected agreement measure in biomedical, psychological, educational, and social-science applications (Cohen 1960). For binary data, however, Cohen's kappa is mathematically identical to the Heidke skill score, proposed in 1926 and still widely used in meteorological forecast verification (Heidke 1926). Moreover, Doolittle had already formulated the same coefficient in 1887 in the context of a dispute over tornado forecasting, noting that in highly unbalanced settings, such as rare diseases or tornadoes, high accuracy can be achieved simply by always predicting the most common outcome (Doolittle 1887). As Stigler's law of eponymy observes, "No scientific discovery is named after its original discoverer" (Stigler 1980).

To address this fragmentation, we introduce `catmetrics`, a new Stata command that computes measures of association, similarity, agreement, information and evaluates deterministic and probabilistic forecasts for categorical data. The command supports dichotomous and polytomous outcomes, as well as nominal and ordinal categories. It reports 301 measures drawn from multiple disciplines, together with their aliases, attainable ranges, neutral values, and directions of improvement. It also returns class-specific quantities and their macro and weighted averages. Thus, `catmetrics` is designed not only as a comprehensive computational tool for applied empirical work but also as a bridge across literatures that use different names, definitions, and notations for algebraically equivalent measures.

2 The `catmetrics` command

The `catmetrics` command accepts inputs in three forms: two observed categorical variables; an observed categorical variable together with predicted probabilities for each category; or a pre-existing $K \times K$ contingency table. The command computes 225 binary-only measures, 63 multicategory nominal or ordinal measures, and 13 multicategory ordinal-only measures.

Measures for probabilistic forecasts

When predicted probabilities are supplied, `catmetrics` computes strictly proper scoring rules and diagnostic measures for probabilistic forecast evaluation. Let Y_i denote the observed outcome of a categorical dependent variable y_i $(i = 1, 2, ..., n)$ that can take values $k \in \{1, ..., K\}$. Let $\delta_{ik} = 1$ if $Y_i = k$, and $\delta_{ik} = 0$ otherwise. Let $\Pr(y_i = k)$ denote the forecasted probability assigned to category k for observation i.

For example, the power score (Good 1971), a strictly proper scoring rule for probabilistic forecasts, is computed as

$$\frac{1}{n}\sum_{i=1}^{n}\left[\frac{1}{\beta} - \sum_{k=1}^{K}\delta_{ik}\Pr(y_i = k)^{\beta-1} + \frac{\beta-1}{\beta}\sum_{k=1}^{K}\left[\Pr(y_i = k)\right]^{\beta}\right].$$

The two-alternative forced choice score (Mason and Weigel 2009), a discrimination score for probabilistic forecasts, is computed as

$$\frac{\sum_{l=1}^{K}\sum_{k\neq l}^{K}\sum_{i=1}^{n+k}\sum_{j=1}^{n+l} I\left[p_{k,i}(l), p_{l,j}(l)\right]}{\sum_{l=1}^{K}\sum_{k\neq l}^{K} n_{+k}n_{+l}},$$

where $p_{k,i}(l)$ is the forecast probability assigned to category l for observation i whose observed category is k, and

$$I\left[p_{k,i}(l), p_{l,j}(l)\right] = \begin{cases} 0 & \text{if} \quad p_{l,j}(l) < p_{k,i}(l) \\ 0.5 & \text{if} \quad p_{l,j}(l) = p_{k,i}(l) \\ 1 & \text{if} \quad p_{l,j}(l) > p_{k,i}(l) \end{cases}.$$

After computing the probability-based measures, `catmetrics` converts the predicted probabilities into a hard classification using the maximum-probability rule: the predicted category is assigned to the category with the largest forecasted probability.

Measures based on contingency tables

Using either two observed categorical variables or an observed categorical variable paired with predicted probabilities, `catmetrics` constructs a contingency table, also known as a cross-tabulation, or a cross-classification table. In a forecasting context, this table is a confusion matrix, with one variable representing the realized category and the other representing the predicted category. The table displays the joint frequency distribution of the two variables and forms the basis for all measures of association, similarity, agreement, information and deterministic forecast-verification computed by the command.

A general $K \times K$ table for variables x and y is shown in Table 1.

Table 1. KxK contingency table (confusion matrix)

	$x = 1$	...	$x = K$	Total
$y = 1$	n_{11}	...	n_{1K}	$n_{1+} = \sum_{j=1}^{K} n_{1j}$
...	...	...	...	...
$y = K$	n_{K1}	...	n_{KK}	$n_{K+} = \sum_{j=1}^{K} n_{Kj}$
Total	$n_{+1} = \sum_{i=1}^{K} n_{i1}$	...	$n_{+K} = \sum_{i=1}^{K} n_{iK}$	$n = \sum_{i=1}^{K} \sum_{j=1}^{K} n_{ij}$

In asymmetric applications, x is the antecedent, earlier, or explanatory variable and y is the consequent, later, or dependent variable. In the context of forecasting, x denotes the realized outcomes and y denotes the predicted ones. Thus, columns represent actual outcomes and rows represent forecasts, and n is the total number of observations.

All deterministic categorical measures computed by `catmetrics` are functions of the entries in this table. For example, the Stuart τ_c coefficient (Stuart 1953) is computed as

$$\frac{2K[\sum_{i=1}^{K}\sum_{j=1}^{K} n_{ij}(\sum_{h>i}\sum_{k>j} n_{hk}) - \sum_{i=1}^{K}\sum_{j=1}^{K} n_{ij}(\sum_{h>i}\sum_{k<j} n_{hk})]}{n^2(K-1)}$$

Class-specific measures

For binary data, category 1 is treated as the positive or event class, and category 2 is treated as the negative or non-event class. Many binary association and forecast-evaluation measures do not treat the positive and negative categories symmetrically. This asymmetry is natural in many applications. Measures developed for event prediction, such as tornado versus no tornado, disease versus no disease, success versus failure, or presence versus absence, often give greater weight to occurrences than to non-occurrences. In addition, joint absences do not always provide evidence of similarity between two objects. Consequently, true positives (n_{11}) and true negatives (n_{22}) are often treated differently. False positives (n_{12}) and false negatives (n_{21}) may also have different costs and consequences.

For example, the G-measure, also known as the Driver–Kroeber index or the Fowlkes–Mallows index, is defined as

$$G\text{-measure} = \frac{n_{11}}{\sqrt{n_{1+}n_{+1}}} = \sqrt{\frac{n_{11}}{n_{1+}}\frac{n_{11}}{n_{+1}}} = \sqrt{\text{Precision} \times \text{Recall}}$$

This is the geometric mean of precision ($\frac{n_{11}}{n_{1+}}$) and recall $\frac{n_{11}}{n_{+1}}$, also known as sensitivity or the true positive rate. The G-measure ignores true negatives (n_{22}) and evaluates performance on the positive class, penalizing poor precision and poor recall symmetrically.

In some applications, however, the negative class is substantively important rather than merely a background category. For example, in ecology, showing that two species are consistently absent from the same environments may be as meaningful as showing that they co-occur. In such cases, researchers may shift attention to the negative class and compute the negative-class G-measure,

$$\text{Negative-class } G\text{-measure} = \frac{n_{22}}{\sqrt{n_{2+}n_{+2}}} = \sqrt{\frac{n_{22}}{n_{2+}}\frac{n_{22}}{n_{+2}}} = \sqrt{\text{NPV} \times \text{TNR}},$$

where NPV is the negative predictive value and TNR is the true negative rate, or specificity.

The command computes both positive-class and negative-class versions for all 225 binary measures. Note that not all binary measures are class-specific: some are invariant to the permutation of the two categories. For example, the Heidke skill score, $\frac{2(n_{11}n_{22}-n_{12}n_{21})}{n_{+1}n_{2+}+n_{1+}n_{+2}}$, is unchanged when the positive and negative labels are exchanged.

Class-specific evaluation for multiclass data

Although many measures are available for multicategory data (`catmetrics` computes 76 such measures), most categorical association and forecast-verification measures were originally developed for binary data and do not always have a multicategory generalization. To apply binary measures in multicategory settings, `catmetrics` uses one-versus-all binarization. This procedure converts a $K \times K$ table into K binary 2×2 tables by treating each class k in turn as the positive class and aggregating all other classes into the negative class.

For each class k, `catmetrics` constructs the following one-versus-all binarized table and computes the class-specific metrics based on each of the 225 binary measures.

Table 2. Class-specific contingency table

	$x = k$	$x \neq k$	Total
$y = k$	$n_{11_{(k)}} = n_{kk}$	$n_{12_{(k)}} = n_{k+} - n_{kk}$	$n_{1+_{(k)}} = n_{k+}$
$y \neq k$	$n_{21_{(k)}} = n_{+k} - n_{kk}$	$n_{22_{(k)}} = n - n_{k+} - n_{+k} + n_{kk}$	$n_{2+_{(k)}} = n - n_{k+}$
Total	$n_{+1_{(k)}} = n_{+k}$	$n_{+2_{(k)}} = n - n_{+k}$	n

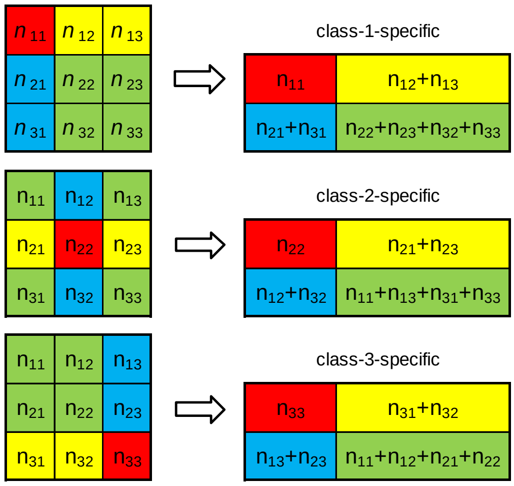
Consider accuracy, the overall proportion of correct predictions: $\frac{1}{n}\sum_{k=1}^{K} n_{kk}$. Although accuracy is useful as a general measure of multicategorical forecast evaluation, a single value can mask substantial differences in class-specific performance. For example, one may ask what proportion of actual observations in each class is correctly predicted. A standard class-specific measure is the hit rate, $\frac{n_{11}}{n_{+1}}$, computed from the appropriate binary table. This measure is also known as the probability of detection in meteorology and forecast verification, recall or the true positive rate in information retrieval, classification, and machine learning, sensitivity in medicine, epidemiology, and diagnostic testing, and power in statistical testing.

For example, consider the following 3×3 confusion matrix:

	$x = 1$	$x = 2$	$x = 3$	Total
$y = 1$	30	9	0	39
$y = 2$	25	163	21	209
$y = 3$	0	9	19	28
Total	55	181	40	276

Figure 1 illustrates the conversion of a 3×3 classification table into three class-specific 2×2 tables.

Figure 1. Conversion of a 3x3 classification table to three 2x2 class-specific tables



The above 3×3 matrix is converted into the following three class-specific binary matrices:

	Class-1-specific	
	$x = 1$	$x \neq 1$
$y = 1$	30	9
$y \neq 1$	25	212

	Class-2-specific	
	$x = 2$	$x \neq 2$
$y = 2$	163	46
$y \neq 2$	18	49

	Class-3-specific	
	$x = 3$	$x \neq 3$
$y = 3$	19	9
$y \neq 3$	21	227

The hit rate for class k is, according to Table 2, $\frac{n_{11(k)}}{n_{+1(k)}} = \frac{n_{kk}}{n_{+k}}$.

In the example, overall accuracy is $\frac{1}{n}\sum_{k=1}^{K} n_{kk} = 0.77$. The class-specific hit rates for classes 1, 2 and 3 are, respectively, $\frac{30}{30+25} = 0.55$, $\frac{163}{163+18} = 0.90$, and $\frac{19}{19+21} = 0.47$. Although class 2 has the highest hit rate, it also has the worst adjusted noise-to-signal ratio. Following Kaminsky and Reinhart (1999), the adjusted noise-to-signal ratio for a binary table is $\frac{n_{12}n_{+1}}{n_{+2}n_{11}}$. For class k, this becomes, according to Table 2, $\frac{n_{12(k)}n_{+1(k)}}{n_{+2(k)}n_{11(k)}}$. In the example, the adjusted noise-to-signal ratios are 0.07, 0.54, and 0.08 for classes 1, 2, and 3, respectively.

To summarize class-specific results and provide a general evaluation, `catmetrics` reports both macro and weighted averages of class-specific measures M_k. The macro average is the unweighted arithmetic mean, $\frac{1}{K}\sum_{k=1}^{K} M_k$, whereas the weighted average is the mean weighted by class frequencies, $\sum_{k=1}^{K} M_k \frac{n_{+k}}{n}$. Macro averages give equal weight to rare and common classes. Weighted averages give greater weight to common classes.

Interpretation of ranges

The command reports attainable ranges for all measures for given n and K, identifies neutral values, such as values corresponding to no skill or statistical independence, and indicates the direction of improvement using arrows. These arrows should be interpreted as general guidelines:

If two arrows point in the same direction, as in $[-1 \rightarrow 0 \rightarrow 1]$, they indicate a progression from perfect negative association to independence and then to perfect positive association.

If arrows point toward a central value, as in $[-1 \rightarrow 0 \leftarrow 1]$, the midpoint is a a neutral value, such as no dependence or no skill.

If only one arrow is shown, as in $[0 \rightarrow 1]$, the arrow points toward stronger positive association or better performance.

If no arrow is shown, as in [0 (never), 1 (always)], none of these directional interpretations applies to the measure's range.

The theoretical range of a measure may depend on the table dimensions and sample size. Consequently, nominal endpoints may not always be attainable, even when marginal totals are not fixed.

Syntax

The command has four syntax forms:

`catmetrics` *var*1 *var*2 `[,` *options*`]`

where *var*1 and *var*2 are two categorical numeric variables with the same number of observed categories. In asymmetric applications, *var*1 precedes *var*2 chronologically or causally. In forecasting applications, *var*1 represents the actual outcome, whereas *var*2 represents the predicted outcome. For binary *var*1 and *var*2, the positive category must be coded using the smallest numeric value.

`catmetrics` *var*1 `[,` *probs*(*varlist*) *options*`]`

where *var*1 contains the realized values of a categorical variable, and *probs*(*varlist*) supplies the predicted probabilities for each category. The probabilities should be listed in ascending category order. The number of supplied probability variables must equal the number of distinct observed categories. If *var*1 is binary, the positive category must be coded using the smallest numeric value.

`catmetrics` *var*1 `[,` *predict options*`]`

where *var*1 again contains the realized outcomes and predicted probabilities are obtained from the most recently estimated Stata model (such as `logistic`, `logit`, `probit`, `oprobit`, `cmclogit`, `mlogit`, `mprobit`, `ziologit`, `zioprobit`) using Stata's post-estimation command `predict`. If *var*1 is binary, the positive category is assumed to be coded using the smallest numeric value.

`catmetrics [,` *mat*(*string*) *options*`]`

where `mat(`*string*`)` supplies a contingency table or confusion matrix directly. If `mat()` is binary, the positive category must be in the first column/row.

In each case, `catmetrics` displays a classification table (confusion matrix), computes the 283 measures of association, agreement, information, similarity, and deterministic categorical forecast evaluation, and returns the class-specific measures for each class with their macro and weighted averages. When predicted probabilities are supplied, `catmetrics` additionally computes 6 strictly proper scoring rules and 12 diagnostic measures for probabilistic forecast evaluation. All computed measures can be exported to an Excel file.

The following options are available.

Options

`excel(`*string*`)` specifies the name of the saved *.xlsx file with all computed measures. By default, the file is named Catmetrics.xlsx.

`noexcel` suppresses creation of the .xlsx file. By default, results are saved to the Catmetrics.xlsx file.

`metrics(`*string*`)` specifies the measures to display in the Results window. Measures can be selected in any order using their indices; see the list of available measures below.

`metrics(`*all*`)` displays all available measures. Probability-based measures are computed only when predicted probabilities are supplied.

`metrics(ordinal)` displays only the measures for ordinal data.

`metrics(`*none*`)` suppresses the display of all measures.

If `metrics()` is not specified, `catmetrics` displays the default set of measures:

for binary data: [1], [2], [9], [42], [64], [65], [119], [120], [125], [132], [134], [146], [148], [163], [205], [212], [234], [241], [256], [280], [281], and [282], and additionally, if predicted probabilities are supplied, [P1], [P2], [P6], [P7], [P8], [P12], [P14], [P15], and [P16];

for multicategory data with $K \geq 3$: [1], [9], [42], [44], [52], [132], [134], [138], [163], [205], [234], [241], [256], [260], and additionally, if predicted probabilities are supplied, [P1], [P2], [P6], [P12], [P14], and [P17].

`classmetrics` displays class-specific measures for binary data. By default, class-specific measures for binary data are not displayed. For multicategory data ($K \geq 3$), if binary metrics are selected in `metrics()`, their class-specific values are always displayed.

`bkappa(`*scalar*`)` specifies the value of κ, a positive irrational number, used for [16] Baulieu coefficient #22. By default, $\kappa = e$.

`crlambda(`*scalar*`)` specifies the value of λ used for [53] Cressie-Read power divergence statistic. By default, $\lambda = 2/3$.

`ealpha(`*scalar*`)` specifies the value of $0 \leq \alpha \leq 1$ used for [271] van Rijsbergen effectiveness E-measure. By default, $\alpha = 0.5$.

`eceq(`*scalar*`)` specifies the value of $q > 0$ used for [P9] expected calibration error. By default, $q = 1$.

`ecem(`*scalar*`)` specifies the integer value of $m \in \{2, 3, ...\}$ used for [P9] expected calibration error. By default, $m = 10$.

`fbeta(`*scalar*`)` specifies the value of $\beta \geq 0$ used for [78] F_β-score. By default, $\beta = 1.5$.

`gaink(`*scalar*`)` specifies the value of $0 < k < 1$ used for [P10] gain at k. By default, $k = 0.1$.

`gfholderp(`*scalar*`)` specifies the value of $p \neq 0$ used for [99-101] generalized Fleiss coefficients (Hölder) #1, #2, and #3. By default, $p = 0.5$.

`gflehmerp(`*scalar*`)` specifies the value of p used for [105-107] generalized Fleiss coefficients (Lehmer) #1, #2, and #3. By default, $p = 1.5$.

`gltheta(`*scalar*`)` specifies the value of $\theta > 0$ used for [139] Gower-Legendre S_θ index. By default, $\theta = 1.5$.

`goalpha(`*scalar*`)` specifies the value of $\alpha > 0$ used for [140] Gray-Orlowska interestingness index. By default, $\alpha = 2$.

`gobeta(scalar)` specifies the value of $\beta > 0$ used for [140] Gray-Orlowska interestingness index. By default, $\beta = 2$.

`ibaalpha(`*scalar*`)` specifies the value of α ($0 \leq \alpha \leq 1$) used for [117] generalized index of balanced accuracy. By default, $\alpha = 0.5$.

`ibam(`*scalar*`)` specifies the value of m, the index of the binary measure, used for [117] generalized index of balanced accuracy. By default, $m = 125$, corresponding to the G-mean.

`kweights(`*string*`)` specifies the weighting matrix used for [274] weighted κ coefficient. The weighting matrix must be symmetric and have the same dimensions as the contingency table. All main diagonal entries ω_{ii} must equal 1, and all off-diagonal entries ω_{ij}, $i \neq j$ must lie in the interval $[0, 1]$. By default, $\omega_{ii} = 1$ and $\omega_{ij} = 0.5$, $i \neq j$.

`liftk(`*scalar*`)` specifies the value of $0 < k < 1$ used for [P11] lift at k. By default, $k = 0.1$.

`mceq(`*scalar*`)` specifies the value of $q > 0$ used for [P13] maximum calibration error. By default, $q = 1$.

`mcem(`*scalar*`)` specifies the integer value of $m \in \{2, 3, ...\}$ used for [P13] maximum calibration error. By default, $m = 10$.

`power(`*scalar*`)` specifies the value of $\beta > 1$ used for [P3] power score. By default, $\beta = 1.5$.

`pseudospherical(`*scalar*`)` specifies the value of $\beta > 1$ used for [P4] pseudospherical score. By default, $\beta = 1.5$.

`sokalw(`*scalar*`)` specifies the value of $w > 0$ used for [244] Sokal-Sneath coefficient #6. By default, $w = 3$.

`talpha(`*scalar*`)` specifies the value of $\alpha > 0$ used for [266] Tversky index. By default, $\alpha = 1/3$.

`tbeta(`*scalar*`)` specifies the value of $\beta > 0$ used for [266] Tversky index. By default, $\beta = 1/3$.

List of available measures

The list of computed measures with their alternative names is adopted from Sirchenko (2026). See Sirchenko (2026) for defining formulas, cross-references, bibliography and further details. Measures suitable only for binary data ($K = 2$) are marked [B], measures suitable for multicategory data ($K \geq 2$) are marked [M], and measures suitable only for ordinal data ($K \geq 2$) are marked [O].

- **Strictly proper scoring rules for probabilistic forecasts**

These measures evaluate the entire forecast probability distribution. They are normalized to be nonnegative and negatively oriented: lower values indicate better forecast, and the perfect value is zero.

P1 Brier score (aliases: half-Brier score, mean squared (probability) error, probability score, quadratic score) [M]
P2 Logarithmic score (aliases: cross-entropy loss, ignorance score, log loss, negative log likelihood / predictive density, surprisal) [M]
P3 Power score (with a user-defined parameter $\beta > 1$) (aliases: generalized quadratic score) [M]
P4 Pseudospherical score (with a user-defined parameter $\beta > 1$) (aliases: generalized spherical score) [M]
P5 Ranked probability score (aliases: Epstein ranked score) [O]
P6 Spherical score [M]

- **Diagnostic measures for probabilistic forecasts**

These measures use the supplied probabilities as inputs but do not evaluate the entire forecast distribution as strictly proper scoring rules do. Instead, they either isolate a specific diagnostic aspect of forecast performance, such as discrimination, ranking, calibration, lift, or gain, or evaluate only a particular functional of the forecast distribution, such as its mode.

P7 Accuracy rate (aliases: correct classification / success rate, fraction / percent / proportion correct, modal / top-1 accuracy, zero-one reward / utility; complement: error rate, misclassification / top-1 error, proportion misclassified, zero-one loss / score) [M]
P8 Average precision (AP) (aliases: non-interpolated area under the precision–recall curve, AUPRC, AUC-PR, PR-AUC) [B]
P9 Expected calibration error (ECE) (with user-defined parameters $q > 0$ and $m \in \{2, 3, ...\}$) [B]
P10 Gain at k (with a user-defined parameter $0 < k < 1$) (aliases: capture rate, (cumulative) gain, gain@k, top-k recall) [B]
P11 Lift at k (with a user-defined parameter $0 < k < 1$) (aliases: (cumulative) lift, lift@k) [B]
P12 Macro average mean probability rate (MAPR) (aliases: macro soft accuracy) [M]
P13 Maximum calibration error (MCE) (with user-defined parameters $q > 0$ and $m \in \{2, 3, ...\}$) [B]
P14 Mean probability rate (MPR) (aliases: mean linear score, mean true class probability, soft accuracy; the complement: mean absolute error) [M]
P15 Normalized area under cumulative accuracy profile (CAP) (aliases: accuracy ratio, AUCAP, normalized area under cumulative gains/lift curve (AUCG, AUGC), Gini index) [B]
P16 Normalized area under receiver operating characteristic (ROC) curve (aliases: AUC, AUROC, concordance c statistics, normalized Wilcoxon–Mann–Whitney U statistic, ROC area, ROC AUC) [B]
P17 Two-alternative forced choice (2AFC) score #1 (aliases: generalized discrimination score) [M]
P18 Two-alternative forced choice (2AFC) score #2 [O]

- **Measures of association, similarity, and deterministic forecast evaluation**

These hard-classification measures are based only on a contingency table or confusion matrix. To avoid redundancy, `catmetrics` generally does not report complementary statistics that sum to one, such as a similarity coefficient and its corresponding dissimilarity coefficient. However, several common complements with distinct and widely used names are reported. Examples include: accuracy $\frac{1}{n}\sum_{k=1}^{K} n_{kk}$ and error rate $\frac{1}{n}\sum_{i=1}^{K}\sum_{j\neq i} n_{ij}$, precision $\frac{n_{11}}{n_{1+}}$ and false alarm ratio $\frac{n_{12}}{n_{1+}}$, hit rate $\frac{n_{11}}{n_{+1}}$ and false negative rate $\frac{n_{21}}{n_{+1}}$, predicted positive rate $\frac{n_{1+}}{n}$ and predicted negative rate $\frac{n_{2+}}{n}$, negative predicted value $\frac{n_{22}}{n_{2+}}$ and false omission rate $\frac{n_{21}}{n_{2+}}$, specificity $\frac{n_{22}}{n_{+2}}$ and false positive rate $\frac{n_{12}}{n_{+2}}$.

1 Accuracy (aliases: (crude / observed/ raw) agreement, causal support, classification accuracy/rate, count R^2, fraction/percent/proportion correct, hit score, Holsti coefficient of agreement *C.R.*, Kendall coefficient, Osgood coefficient, Rand index, ratio test discriminant, simple matching coefficient, Sokal-Michener coefficient, success rate) [M]
2 Added value (aliases: absolute Quetelet index, centered confidence, change of support, gain, Pavillon index) [B]
3 Adjusted noise-to-signal ratio [B]
4 Alroy coefficient (aliases: corrected Forbes F coefficient) [B]
5 Anderberg coefficient [B]
6 Anderberg D similarity coefficient [B]
7 Appleman index [B]
8 Atkinson similarity [M]
9 Balanced accuracy (aliases: balanced classification rate; complement: balanced error rate) [M]
10 Balanced error rate (aliases: balanced classification error / rate, half total error rate, complement: balanced accuracy) [M]
11 Baroni-Urbani–Buser coefficient #1 [B]
12 Baroni-Urbani–Buser coefficient #2 [B]
13 Base rate (aliases: event probability, sample climate, (true) prevalence) [B]
14 Batagelj-Bren distance Q_0 (aliases: inverse (diagnostic) odds ratio) [B]
15 Baulieu dissimilarity coefficient #13 [B]
16 Baulieu dissimilarity coefficient #22 (with a user-supplied parameter κ, a positive irrational number) [B]
17 Baulieu dissimilarity coefficient #23 [B]
18 Baulieu dissimilarity coefficient #24 [B]
19 Baulieu dissimilarity coefficient #25 [B]
20 Baulieu dissimilarity coefficient #27 [B]
21 Baulieu dissimilarity coefficient #28 [B]

22 Baulieu dissimilarity coefficient #29 [B]
23 Baulieu dissimilarity coefficient #30 [B]
24 Baulieu dissimilarity coefficient #31 [B]
25 Baulieu dissimilarity coefficient #32 [B]
26 Baulieu dissimilarity coefficient #33 [B]
27 Benini $\phi/\phi_{\max}$ correlation coefficient #1 [B]
28 Benini $\phi/\phi_{\max}$ correlation coefficient #2 (aliases: Benini attraction/repulsion, Gini coefficient, bound-adjusted Cohen κ) [M]
29 Benini correlation coefficient #3 (aliases: Benini attraction, Cole coefficient, Köppen index, certainty factor) [B]
30 Benini correlation coefficient #4 (aliases: Benini attraction, Cole coefficient) [B]
31 Bennet S coefficient (aliases: Brennan-Prediger κ_n coefficient, Guilford G coefficient, Janson-Vegelius C coefficient, Maxwell random error RE, Perreault-Leigh I_r index, reliability coefficient) [M]
32 Berger-Parker dominance index (aliases: proportional dominance) [M]
33 Bias index (aliases: assessment / forecast / frequency / response bias) [B]
34 Blaheta-Johnson unigram subtuples measure [B]
35 Braun-Blanquet similarity index [B]
36 Bray-Curtis dissimilarity index (aliases: Whittaker index) [M]
37 Brin conviction coefficient [B]
38 Causal confidence [B]
39 Causal confirmed confidence [B]
40 Causal confirm [B]
41 Chord distance metric [B]
42 Clayton skill score (aliases: Δp, index of separation, markedness) [M]
43 Clement reliability coefficient [B]
44 Cohen κ coefficient (aliases: Cohen agreement, Cohen kappa, inter-rater agreement, inter-rater reliability, kappa coefficient) [M]
45 Cole C_5 correlation coefficient [B]
46 Collective strength index [B]
47 Consonni–Todeschini similarity index #1 [B]
48 Consonni–Todeschini similarity index #2 [B]
49 Consonni–Todeschini similarity index #3 [B]
50 Consonni–Todeschini similarity index #4 [B]
51 Consonni–Todeschini similarity index #5 [B]
52 Cramér concordance coefficient [M]
53 Cressie-Read power divergence statistic (with a user-supplied parameter λ; identical to: Neyman χ^2 test statistic if $\lambda = -2$; Freeman-Tukey (F^2 or T^2) test statistic if $\lambda = -1/2$; likelihood ratio G^2 test statistic if $\lambda \to 0$; modified likelihood ratio test statistics if $\lambda = -1$, Pearson χ^2 if $\lambda = 1$; optimal Cressie-Read power divergence statistic if $\lambda = 2/3$) [M]
54 Czekanowski index (aliases: Burt coefficient, Dice coefficient, Gleason index, F_1-score, harmonic mean of precision and recall, Nei-Li index, positive specific agreement, proportion of specific agreement, Sørensen index, Sørensen–Dice coefficient, Tversky index (special case if $\alpha = \beta = 1/2$), Upholt F coefficient, Zijdenbos index; complement: Bray-Curtis index, Lance–Williams coefficient) [B]
55 Dennis z-score (aliases: z-score) [B]
56 Dependency measure [B]
57 Descriptive confirm [B]
58 Digby correlation coefficient [B]
59 Discriminant power [B]
60 Discrimination d' distance (aliases: signal detection sensitivity d') [B]
61 Dominance index [B]
62 Donaldson bias index [B]
63 Doolittle association ratio (aliases: Cramér coefficient, inertia, mean square contingency φ^2) [M]
64 Doolittle raw accuracy (aliases: correlation ratio, Doolittle H_u, Michelet index, Sorgenfrei coefficient, unbiased hit rate H_u, Wagner index; complement: Baulieu coefficient #12) [B]
65 Driver–Kroeber similarity index (aliases: Fowlkes–Mallows index, G-measure, geometric mean of recall and precision, cosine similarity, Ochiai coefficient, Otsuka coefficient) [B]
66 Error rate (aliases: (mis)classification error, misclassification rate, zero-one loss; the complement: accuracy) [M]

67 Example and counterexample rate [B]
68 Extremal dependence index [B]
69 Extreme dependency index [B]
70 Eyraud index [B]
71 Fager–McGowan similarity index #1 (aliases: corrected Ochiai/cosine similarity) [B]
72 Fager–McGowan similarity index #2 [B]
73 Faith similarity index [B]
74 False alarm ratio (aliases: false discovery rate; complement: precision) [B]
75 False negative rate (aliases: miss rate; complement: hit rate) [B]
76 False omission rate (aliases: detection failure ratio, miss ratio; complement: negative predicted value) [B]
77 False positive rate (aliases: fall-out, false alarm rate, probability of false detection; type II error rate; complement: specificity) [B]
78 F_β-score (aliases: weighted harmonic mean of the precision and hit rate; complement: van Rijsbergen effectiveness E measure) [B]
79 Fisher exact statistic (aliases: Freeman-Halton statistic) [M]
80 Fleiss–Levin–Paik agreement index (aliases: negative-class Dice, harmonic mean of negative predictive value and specificity, negative-class F_1-score, negative percent agreement, negative (specific) agreement, proportion of negative agreement, negative-class Sørensen) [B]
81 Forbes coefficient #1 (aliases: Forbes F coefficient, independence, interest, Kocher–Wong measure, lift) [B]
82 Forbes coefficient #2 (aliases: Benini correlation/repulsion, Loevinger H coefficient, relative improvement over chance (RIOC)) [B]
83 Fossum index [B]
84 Freeman-Tukey (F^2 or T^2) statistic [M]
85 Freeman-Tukey test (F^2 or T^2) statistic (asymptotic approximation) (identical to Cressie-Read power divergence statistic if $\lambda = -1/2$) [M]
86 F-score adjusted [B]
87 Galton agreement coefficient (aliases: Cole C_7 coefficient, conditional kappa, Light coefficient) [B]
88 Ganascia coefficient (aliases: confirmed confidence) [B]
89 Generalized Fleiss correlation coefficient (arithmetic) #2 [B]
90 Generalized Fleiss correlation coefficient (contraharmonic) #1 [B]
91 Generalized Fleiss correlation coefficient (contraharmonic) #2 [B]
92 Generalized Fleiss correlation coefficient (contraharmonic) #3 [B]
93 Generalized Fleiss correlation coefficient (harmonic) #1 (identical to original Fleiss coefficient) [B]
94 Generalized Fleiss correlation coefficient (harmonic) #2 [B]
95 Generalized Fleiss correlation coefficient (harmonic) #3 [B]
96 Generalized Fleiss correlation coefficient (Heronian) #1 [B]
97 Generalized Fleiss correlation coefficient (Heronian) #2 [B]
98 Generalized Fleiss correlation coefficient (Heronian) #3 [B]
99 Generalized Fleiss correlation coefficient (Hölder) #1 (with a user-defined parameter $p \neq 0$) [B]
100 Generalized Fleiss correlation coefficient (Hölder) #2 (with a user-defined parameter $p \neq 0$) [B]
101 Generalized Fleiss correlation coefficient (Hölder) #3 (with a user-defined parameter $p \neq 0$) [B]
102 Generalized Fleiss correlation coefficient (identric) #1 [B]
103 Generalized Fleiss correlation coefficient (identric) #2 [B]
104 Generalized Fleiss correlation coefficient (identric) #3 [B]
105 Generalized Fleiss correlation coefficient (Lehmer) #1 (with a user-defined parameter p) [B]
106 Generalized Fleiss correlation coefficient (Lehmer) #2 (with a user-defined parameter p) [B]
107 Generalized Fleiss correlation coefficient (Lehmer) #3 (with a user-defined parameter p) [B]
108 Generalized Fleiss correlation coefficient (logarithmic) #1 [B]
109 Generalized Fleiss correlation coefficient (logarithmic) #2 [B]
110 Generalized Fleiss correlation coefficient (logarithmic) #3 [B]
111 Generalized Fleiss correlation coefficient (quadratic) #1 [B]
112 Generalized Fleiss correlation coefficient (quadratic) #2 [B]
113 Generalized Fleiss correlation coefficient (quadratic) #3 [B]
114 Generalized Fleiss correlation coefficient (Seiffert) #1 [B]
115 Generalized Fleiss correlation coefficient (Seiffert) #2 [B]
116 Generalized Fleiss correlation coefficient (Seiffert) #3 [B]
117 Generalized index of balanced accuracy (IBA) (with user-defined parameters $0 < \alpha < 1$ and M - an index of any binary metric) [B]

118 Gerrity skill score (aliases: special case of Gandin-Murphy equitable skill score) [O]
119 Gilbert index (aliases: critical success index, Driver-Kroeber coefficient, intersection over union, Jaccard index, Jaccardized Czekanowski index, mutual support, ratio of verification, Ružička index, Sneath index, Tanimoto coefficient, threat score, Tversky index) [B]
120 Gilbert skill score (aliases: equitable threat score, ratio of success) [B]
121 Gilbert–Wells index [B]
122 Gini coefficient #1 [B]
123 Gini coefficient #2 [B]
124 Gini impurity index (aliases: Gibbs M1 index, Gini–Simpson index, index of differentiation, Simpson diversity index) [M]
125 G-mean (aliases: geometric mean of sensitivity and specificity) [B]
126 G-mean adjusted (aliases: adjusted geometric mean) [B]
127 Goodall index (aliases: Austin–Colwell index) [M]
128 Goodman concomitance coefficient [B]
129 Goodman–Kruskal coefficient #1 [B]
130 Goodman–Kruskal coefficient #2 [B]
131 Goodman-Kruskal γ coefficient [O]
132 Goodman-Kruskal λ coefficient (aliases: Guttman λ coefficient) [M]
133 Goodman-Kruskal λ_r coefficient (aliases: adjusted count R^2, Brennan-Prediger κ_b coefficient) [M]
134 Goodman-Kruskal τ coefficient [M]
135 Goodman-Kruskal weighted λ coefficient [M]
136 Goodman unweighted association index [B]
137 Goodman weighted association index [B]
138 Gorodkin R_K correlation coefficient (aliases: generalized Matthews correlation, generalized Yule φ, Gini index of homophyly) [M]
139 Gower-Legendre S_θ index (with user-defined parameter $\theta > 0$) (aliases: parameterized overall agreement; identical to Sokal-Sneath #1 when $\theta = 1/2$, to accuracy when $\theta = 1$, to Rogers–Tanimoto when $\theta = 2$) [M]
140 Gray–Orlowska interestingness index (with user-defined parameters $\alpha > 0$ and $\beta > 0$) [B]
141 Grier B' bias index [B]
142 Guttman coefficient [B]
143 Hamann coefficient (aliases: Hubert Γ statistic) [M]
144 Harris–Lahey index [B]
145 Hawkins–Dotson coefficient [B]
146 Heidke skill score (aliases: binary Cohen κ, Doolittle index, generalized Fleiss correlation coefficient (arithmetic) #3, κ index of agreement) [B]
147 Hellinger distance [B]
148 Hit rate (aliases: completeness, Dice coefficient, fraction correct, power, prefigurance, probability of detection, producer's accuracy, positive percent agreement, recall, sensitivity, true positive rate) [B]
149 Höffding coefficient #1 [M]
150 Höffding coefficient #2 [M]
151 Hubert–Arabie adjusted Rand index [M]
152 Index of dissimilarity (aliases: analyzing method patterns to locate errors (AMPLE), Duncan dissimilarity / segregation index) [B]
153 Information quality ratio [B]
154 J-measure (aliases: information content) [B]
155 Johnson (aliases: sum of confidences) [B]
156 Kendall τ_a correlation coefficient [O]
157 Kendall τ_b correlation coefficient with an adjustment for ties [O]
158 Kent–Foster #1 coefficient [B]
159 Kent–Foster #2 coefficient [B]
160 Kitamura-Matsumoto index (aliases: modified Dice coefficient) [B]
161 Klösgen measure [B]
162 Köppen index [O]
163 Krippendorff α coefficient [M]
164 Kuder–Richardson coefficient [B]
165 Kuhns coefficient #1 (aliases: arithmetic mean overlap) [B]
166 Kuhns coefficient #2 (aliases: proportion of overlap above independence) [B]
167 Kulczyński index #1 [B]

168 Kulczyński index #2 (aliases: Driver–Kroeber coefficient) [B]
169 Lakshmanamurti Λ coefficient [B]
170 Laplace correction [B]
171 Least contradiction index [B]
172 Lerman implication index [B]
173 Leverage (aliases: Baulieu coefficient #20, Piatetsky–Shapiro index, weighted relative accuracy) [B]
174 Likelihood ratio G^2 statistic (aliases: (log) likelihood ratio χ^2 statistic, special case of Cressie-Read power divergence statistic if $\lambda \to 0$) [M]
175 Log Forbes measure (aliases: Gilbert-Wells index, log pointwise mutual information) [B]
176 Log frequency biased index [B]
177 Log odds ratio amended statistic [B]
178 Log odds ratio statistic [B]
179 Mak ρ coefficient [B]
180 Mantel-Haenszel statistic (aliases: linear-by-linear association statistic, nonzero correlation statistic) [O]
181 Maron–Kuhns coefficient [B]
182 Maxwell B coefficient (aliases: Maxwell adjusted random error RE, maximum-corrected Bennett S coefficient, bounded uniform agreement index) [M]
183 Maxwell–Pilliner coefficient (aliases: generalized Fleiss coefficient (arithmetic) #1) [B]
184 McConnaughey coefficient [B]
185 Merton correct prediction CP index [B]
186 Michael index [B]
187 Modified likelihood ratio statistics (aliases: minimum discrimination information (MDI) statistic, modified G^2 statistic, special case of Cressie-Read power divergence statistic if $\lambda = -1$) [M]
188 Mountford index [B]
189 Mueller–Schuessler index of qualitative variation (aliases: Gibbs's M2 index, index of qualitative variation (IQV)) [M]
190 Mutual dependency [B]
191 Mutual information statistic (aliases: information gain) [M]
192 Negative likelihood ratio $LR-$ statistic (aliases: logical necessity) [B]
193 Negative predicted value (aliases: inverse precision, true negative accuracy; complement: false omission rate) [B]
194 Neyman modified χ^2 statistic (aliases: special case of Cressie-Read power divergence statistic if $\lambda = -2$) [M]
195 Normalized Google distance (NGD) metric [B]
196 Odds ratio (aliases: diagnostic odds ratio, cross-product ratio) [B]
197 Odds ratio amended [B]
198 Optimized precision [B]
199 P_4-score [B]
200 Pattern difference [B]
201 Pearson χ^2 statistic (aliases: Cramér statistic, special case of power divergence statistic if $\lambda = 1$) [M]
202 Pearson contingency C coefficient [M]
203 Pearson–Heron correlation coefficient (aliases: tetrachoric correlation coefficient) [B]
204 Pearson φ coefficient (aliases: mean square contingency coefficient) [M]
205 Peirce skill score (aliases: (bookmaker) informedness, corrected hit probability, Δp, Gandin–Murphy equitable skill score, Hanssen–Kuipers discriminant, Kuipers skill score, true skill score, true skill statistic, Woodworth index, Youden J statistic) [M]
206 Pietra statistic (aliases: Schutz coefficient) [M]
207 Poisson-Stirling index [B]
208 Pollack-Norman A' sensitivity statistic [B]
209 Pollaczek-Geiringer correlation coefficient [O]
210 Positive likelihood ratio $LR+$ statistic (aliases: Bayes factor, logical sufficiency) [B]
211 Positive matching coefficient [B]
212 Precision (aliases: absolute support, certainty factor, confidence, correct alarm ratio, Dice coefficient, frequency of hits, positive predictive value, post agreement, strength, success ratio; complement: predicted negative rate) [B]
213 Predicted negative rate (aliases: apparent non-prevalence, negative call rate, predicted negative proportion, rejection rate / ratio, test-negative proportion, complement: predicted positive rate)

[B]

214 Predicted positive rate (aliases: apparent/predicted prevalence, applicability, coverage, generality, probability of a forecast of occurrence, (positive) forecast rate/frequency, selection ratio; complement: predicted negative rate) [B]

215 Prevalence threshold index [B]

216 Putative causal dependency index [B]

217 Relative risk ratio (aliases: (adverse) impact ratio (AIR), risk ratio) [B]

218 Relative Quetelet index (aliases: centered lift, lift minus one, varying rates liaison (VRL)) [B]

219 Renkonen similarity index (aliases: percentage similarity index; complement: index of dissimilarity) [B]

220 Replacement component (Jaccard-based) (aliases: Ružička replacement index) [B]

221 Replacement component (Sørensen-based) [B]

222 Replacement component (Braun-Blanquet-based) (aliases: Savage replacement index) [B]

223 Richness difference (Jaccard-based) (aliases: Ružička abundance difference) [B]

224 Richness difference (Sørensen-based) (aliases: abundance difference) [B]

225 Richness difference (Braun-Blanquet based) (aliases: abundance difference) [B]

226 Rogers–Tanimoto coefficient [M]

227 Rogot–Goldberg index [B]

228 Root mean square difference (aliases: normalized Euclidean distance, Sokal distance) [B]

229 Rousseau index [B]

230 Roux index #1 [B]

231 Roux index #2 [B]

232 Russell–Rao coefficient (aliases: support) [B]

233 Schrank index [B]

234 Scott π coefficient (aliases: Scott index of inter-code agreement, the two-rater special case of Fleiss κ) [M]

235 Sebag–Schoenauer index [B]

236 Shape difference [B]

237 Simple matching coefficient [B]

238 Size difference (aliases: Baulieu coefficient #26, Penrose size difference) [B]

239 Sokal–Sneath similarity coefficient #1 (aliases: Anderberg coefficient) [M]

240 Sokal–Sneath similarity coefficient #2 (aliases: Anderberg coefficient) [M]

241 Sokal–Sneath similarity coefficient #3 (aliases: Anderberg coefficient, Rogot–Goldberg index) [M]

242 Sokal–Sneath similarity coefficient #4 (aliases: Gower coefficient, Ochiai coefficient) [B]

243 Sokal–Sneath similarity coefficient #5 (aliases: Anderberg coefficient) [B]

244 Sokal-Sneath similarity coefficient #6 (with a user-supplied parameter $w > 0$) (aliases: weighted Jaccard index; identical to Sokal–Sneath #5 if $w = 1/2$, to Gilbert score if $w = 1$, to Gower-Legendre T_θ if $w = 1/\theta$, to Czekanowski if $w = 2$, to Sørensen if $w = 4$ and to Anderberg if $w = 8$) [B]

245 Somers d statistics [O]

246 Sørensen similarity coefficient [B]

247 Specificity (aliases: inverse recall, selectivity, true negative rate; complement: false positive rate) [B]

248 Standard deviation agreement (SDAI) index (aliases: Armitage–Blendis–Smyllie index) [B]

249 Steffensen Ψ^2 coefficient [M]

250 Steffensen ω coefficient [M]

251 Stiles coefficient [B]

252 Stuart τ_c correlation coefficient (for ordinal variables; aliases: Kendall τ_c coefficient, Kendall-Stuart τ_c coefficient) [O]

253 Success index (aliases: Tarwid index) [B]

254 Szymkiewicz-Simpson similarity coefficient (aliases: confidence, overlap) [B]

255 Tarantula metric [B]

256 Theil uncertainty coefficient U (aliases: entropy, normalized mutual information (NMI) (directional), proficiency, (asymmetric) uncertainty coefficient) [M]

257 Theil symmetric uncertainty coefficient U (aliases: normalized mutual information (NMI) (symmetric), symmetric uncertainty (SU) coefficient) [M]

258 Tönnies coefficient [O]

259 Tschuprow T bias-corrected coefficient (aliases: Cramér V bias-corrected coefficient) [M]

260 Tschuprow T coefficient (aliases: Cramér V coefficient or φ_c) [M]

261 $t-$score [B]

262 Tulloss R cost index [B]
263 Tulloss S cost index [B]
264 Tulloss T combined costs index (aliases: tripartite similarity T index) [B]
265 Tulloss U cost index [B]
266 Tversky index (with user-defined parameters $\alpha > 0$ and $\beta > 0$; identical to Sokal-Sneath #6 if $\alpha = \beta = 1/w$) [B]
267 Two-alternative forced choice $2AFC$ statistic #1 [M]
268 Two-alternative forced choice $2AFC$ statistic #2 [O]
269 Upholt S coefficient [B]
270 Van der Maarel coefficient [B]
271 Van Rijsbergen effectiveness E-measure (with user-defined parameter $0 \leq \alpha \leq 1$) (complement: F_β-score) [B]
272 Variance dissimilarity measure [B]
273 Warrens coefficient (aliases: harmonic mean of precision, hit rate, negative predicted value and specificity) [B]
274 Weighted κ coefficient (with user-supplied parameters $0 \leq \omega_{ij} \leq 1$ with all $\omega_{ii} = 1$ and all $\omega_{ij} = \omega_{ji}$) (aliases: weighted κ, weighted agreement, weighted Cohen kappa, identical to unweighted Cohen κ if $\omega_{ij} = 0, i \neq j$) [M]
275 Woodcock similarity coefficient (complement: Baulieu dissimilarity coefficient #20) [B]
276 Yao–Liu one-way support measure [B]
277 Yao–Liu two-way support measure [B]
278 Yao–Liu two-way support variation [B]
279 Yates χ^2 statistics [B]
280 Yule φ correlation coefficient (aliases: generalized Fleiss correlation coefficient (geometric), Matthews correlation MCC coefficient, Pearson product-moment correlation coefficient φ, r_φ, tetrachoric correlation, Yule correlation) [B]
281 Yule Q coefficient (aliases: Goodman–Kruskal γ coefficient, odds ratio skill score, Yule association) [B]
282 Yule Y coefficient (aliases: Yule colligation, Yule σ) [B]
283 Zhang coefficient [B]

3 Examples

The `catmetrics` command can use a pre-existing $K \times K$ classification table (confusion matrix) as an input. If no measures are specified, the command displays a default set of measures in the Results window. By default, the complete output is also saved to the Excel file Catmetric.xlsx:

```
. matrix C = (11,2,3  3,16,6  7,5,22)

. catmetrics, mat(C)

Contingency Table

    Actual |      1      2      3 |    Total
-----------+----------------------+----------
 Predicted |                      |
         1 |     11      2      3 |       16
         2 |      3     16      6 |       25
         3 |      7      5     22 |       34
-----------+----------------------+----------
     Total |     21     23     31 |       75

Selected measures of association & hard forecast evaluation
-----------------------------------------------------------------------
Measure                                              Range       Value
-----------------------------------------------------------------------
1. Accuracy                                        [0 -> 1]     0.6533
9. Balanced accuracy                        [0 -> 0.5 -> 1]     0.6430
42. Clayton skill score                      [-1 -> 0 -> 1]     0.4766
44. Cohen kappa coefficient                  [-1 -> 0 -> 1]     0.4672
52. Cramer concordance coefficient                 [0 -> 1]     0.2289
132. Goodman-Kruskal lambda coefficient            [0 -> 1]     0.3659
134. Goodman-Kruskal tau coefficient               [0 -> 1]     0.2262
138. Gorodkin Rk coefficient                 [-1 -> 0 -> 1]     0.4697
163. Krippendorff alpha coefficient             [-.99 -> 1]     0.4694
205. Peirce skill score                      [-1 -> 0 -> 1]     0.4629
234. Scott pi coefficient                         [-1 -> 1]     0.4658
```

```
241. Sokal-Sneath similarity coefficient #3    [0 -> .5 -> 1]    0.6506
256. Theil uncertainty coefficient U                 [0 -> 1]    0.1987
260. Tschuprow T coefficient                         [0 -> 1]    0.4784
-----------------------------------------------------------------------

See the Excel file ´Catmetrics.xlsx´ for the complete output.
```

The command can also take two observed categorical variables as input. Specific measures can be selected for display in the Results window, and parameter values can be supplied for measures that contain user-defined parameters:

```
. webuse lbw, clear
(Hosmer & Lemeshow data)

. catmetrics smoke low, metrics(54 196 217 256 266 280 281 282) talpha(0.2) tbeta(0.2)

Contingency Table

      smoke= |    Positive (0)    Negative (1) |    Total
-------------+---------------------------------+----------
        low= |                                 |
Positive (0) |              86              44 |      130
Negative (1) |              29              30 |       59
-------------+---------------------------------+----------
       Total |             115              74 |      189

Selected measures of association & hard forecast evaluation
---------------------------------------------------------------------
Measure                                             Range      Value
---------------------------------------------------------------------
54. Czekanowski index                            [0 -> 1]     0.7020
196. Odds ratio                        [0 -> 1 -> 8742.25]    2.0219
217. Relative risk ratio                   [0 -> 1 -> 188]    1.3459
256. Theil uncertainty coefficient U             [0 -> 1]     0.0207
266. Tversky index (alpha=.2,beta=.2)            [0 -> 1]     0.8549
280. Yule phi correlation coefficient       [-1 -> 0 -> 1]    0.1614
281. Yule Q coefficient                     [-1 -> 0 -> 1]    0.3382
282. Yule Y coefficient                     [-1 -> 0 -> 1]    0.1742
---------------------------------------------------------------------

See the Excel file ´Catmetrics.xlsx´ for the complete output.
```

Besides, an observed categorical variable can be supplied together with predicted probabilities for each category. In this case, `catmetrics` can report probabilistic forecast measures as well as class-specific hard-classification measures and their macro and weighted averages:

```
. webuse sysdsn1, clear
(Health insurance data)

. quietly mlogit insure age male nonwhite i.site

. predict p*
(option pr assumed; predicted probabilities)
(1 missing value generated)

. catmetrics insure, probs(p1 p2 p3) classmetrics ///
> metrics(P1 P2 P6 P14 P15 P16 P17 1 9 44 54 120 138 148 205)
WARNING: 29 observation(s) with missing data were removed.

Confusion Matrix

    Actual |      1      2      3 |    Total
-----------+----------------------+----------
 Predicted |                      |
         1 |    203    132     28 |      363
         2 |     90    145     17 |      252
         3 |      0      0      0 |        0
-----------+----------------------+----------
     Total |    293    277     45 |      615

Selected proper scores for probabilistic forecasts
---------------------------------------------------
Score                   Range            Value
---------------------------------------------------
P1 Brier score         [0 <- 1]         0.2699
P2 Logarithmic score   [0 <- inf]       1.5835
P6 Spherical score     [0 <- 1]         0.3225
```

```
---------------------------------------------

Selected diagnostic measures for probabilistic forecasts
------------------------------------------------------------------------------------------------
Score                             Range       Value    Class    Class    Class    Macro    Wgted
                                                        (1)      (2)      (3)     averg    averg
------------------------------------------------------------------------------------------------
P14 Mean probability rate        [0 -> 1]     0.460
P15 Normalized area under CAP    [0 -> 1]              0.236    0.260    0.300    0.266    0.252
P16 Normalized area under ROC    [0 -> 1]              0.618    0.630    0.650    0.633    0.626
P17 2AFC score #1                [0 -> 1]     0.627
------------------------------------------------------------------------------------------------

Selected measures of association & hard forecast evaluation
-------------------------------------------------------------------------------------------------
Measure                                Range     Value    Class    Class    Class    Macro    Wgted
                                                           (1)      (2)      (3)     averg    averg
-------------------------------------------------------------------------------------------------
1. Accuracy                          [0 -> 1]    0.566
9. Balanced accuracy            [0 -> 0.5 -> 1]  0.405
44. Cohen kappa coefficient      [-1 -> 0 -> 1]  0.187
54. Czekanowski index                [0 -> 1]             0.619    0.548    0.000    0.389    0.542
120. Gilbert skill score       [-.33 -> 0 -> 1]           0.107    0.116    0.000    0.075    0.104
138. Gorodkin Rk coefficient     [-1 -> 0 -> 1]  0.191
148. Hit rate                        [0 -> 1]             0.693    0.523    0.000    0.405    0.566
205. Peirce skill score          [-1 -> 0 -> 1]  0.177
---------------------------------------------------------------------------------------------------
See the Excel file ´Catmetrics.xlsx´ for the complete output.
```

Alternatively, catmetrics can use an observed categorical variable and obtain the corresponding predicted probabilities directly from the most recently estimated Stata model. For ordinal outcomes, the user can request all measures applicable specifically to ordinal data and suppress creation of the Excel output file:

```
. webuse tobacco, clear
(Fictional tobacco consumption data)

. quietly zioprobit tobacco education income i.female age,  ///
> inflate(education income i.parent age i.female i.religion)

. catmetrics tobacco, pred metrics(ordinal)

Confusion Matrix

    Actual |      0      1      2      3 |    Total
-----------+----------------------------+----------
 Predicted |                            |
         0 |   8006    783    458    586 |     9833
         1 |   1166   2941    162      0 |     4269
         2 |    237     82    416     36 |      771
         3 |     60      0     14     53 |      127
-----------+----------------------------+----------
     Total |   9469   3806   1050    675 |    15000

Selected proper scores for probabilistic forecasts
-----------------------------------------------------------
Score                                  Range          Value
-----------------------------------------------------------
P5 Ranked probability score [ORD]    [0 <- 1]        0.0814
-----------------------------------------------------------
ORD: suited for ordinal data only.

Selected diagnostic measures for probabilistic forecasts
---------------------------------------------
Score                       Range        Value
---------------------------------------------
P18 2AFC score #2 [ORD]    [0 -> 1]     0.8167
---------------------------------------------
ORD: suited for ordinal data only.

Selected measures of association & hard forecast evaluation
----------------------------------------------------------------------------
Measure                                                 Range         Value
----------------------------------------------------------------------------
118. Gerrity skill score [ORD]                     [-1 -> 0 -> 1]      0.2869
131. Goodman-Kruskal gamma coefficient [ORD]       [-1 -> 0 -> 1]      0.6397
156. Kendall tau-a coefficient [ORD]             [-.75 -> 0 -> .75]    0.2255
157. Kendall tau-b coefficient [ORD]               [-1 -> 0 -> 1]      0.4440
```

```
162. Koppen index [ORD]                                  [0 -> 1]      0.8358
180. Mantel-Haenszel statistic [ORD]                [0 -> 14999]     1.7e+03
209. Pollaczek-Geiringer coefficient [ORD]       [-1 -> 0 -> 1]      0.5960
245. Somers d statistics [ORD]                   [-1 -> 0 -> 1]      0.4253
252. Stuart tau-c correlation [ORD]              [-1 -> 0 -> 1]      0.3007
258. Tonnies coefficient [ORD]                         [-2 -> 2]      1.2829
268. 2AFC statistic #2 [ORD]                     [0 -> .5 -> 1]      0.7127
-----------------------------------------------------------------------------
ORD: suited for ordinal data only

See the Excel file ´Catmetrics.xlsx´ for the complete output.
```

4 Concluding remarks

Any attempt to summarize association or similarity in a cross-classification, or to evaluate categorical forecasts with a single scalar statistic, necessarily discards information. Because different measures emphasize different aspects of the joint distribution, the choice of one measure over another implicitly determines which information is prioritized and which is ignored. Consequently, no single measure is uniformly optimal across all substantive contexts.

For detailed discussions of the mathematical properties, contextual interpretations, and comparative relationships among these measures, as well as practical guidance on metric selection, researchers are encouraged to consult the methodological literature. Foundational and comparative contributions include, among others, Albatineh et al. (2006), Baulieu (1989), Bickel (2007), Brusco et al. (2021), Choi et al. (2010), Daan (1984), Ebert and Milne (2022), Geng and Hamilton (2006), Gneiting and Raftery (2007), Goodman (2000), Goodman and Kruskal (1959), Gower and Legendre (1986), Hubálek (1982), Jolliffe and Stephenson (2012), Sokal and Sneath (1963), Sokal and Rohlf (1981), Todeschini et al. (2012), and Warrens (2008a,b).

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